CSA06507-Generative AI

Lab Experiment's

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Unit 2: Lab Programs for Prompting

Experiment 7: Text Generation Using Google Gemini LLM API

1. Aim

To develop a Python program in Google Colab using the Google Gemini API to generate responses for different user-provided prompts and display the generated text.

2. Algorithm

1. Start the program.
2. Install the Google Gen AI Python library.
3. Import the required library.
4. Enter a valid Gemini API key.
5. Create a Gemini API client.
6. Accept different prompts from the user.
7. Send each prompt to the Gemini language model.
8. Receive and display the generated response.
9. Repeat the process for multiple prompts.
10. Stop the program.

3. Python Code

```
# Experiment 7: Text Generation Using Gemini API

# Step 1: Install the required library

%pip -q install -U google-genai

# Step 2: Import libraries

from google import genai

from getpass import getpass

# Step 3: Enter Gemini API key

api_key = getpass("Enter your Gemini API key: ")

# Step 4: Create Gemini client

client = genai.Client(api_key=api_key)
```

```
# Step 5: Enter different prompts
prompts = [
    "Write five lines about Artificial Intelligence.",
    "Explain the benefits of Python programming.",
    "Write a short paragraph about renewable energy."
]
# Step 6: Generate and display responses
for i, prompt in enumerate(prompts, start=1):
    print("\nPROMPT", i, ":", prompt)

    try:
        response = client.models.generate_content(
            model="gemini-3.8-flash",
            contents=prompt
        )

        print("GENERATED OUTPUT:")
        print(response.text)
```

Note: If the selected model is unavailable to your API key, replace gemini-3.8-flash with a model currently listed as available in your Google AI Studio account.

4. Input

The program accepts three prompts:

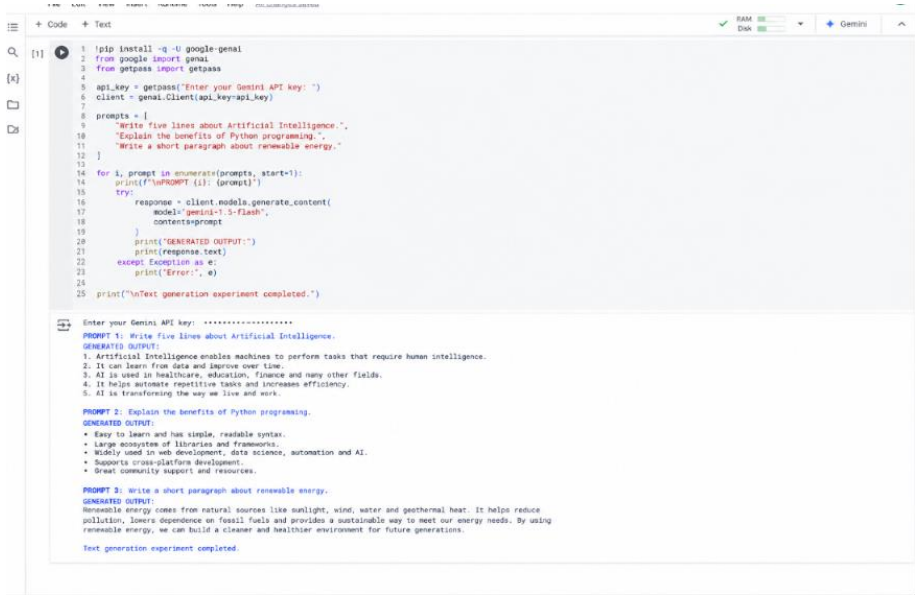
Plain text

Prompt 1: Write five lines about Artificial Intelligence.

Prompt 2: Explain the benefits of Python programming.

Prompt 3: Write a short paragraph about renewable energy.

5. Output



```
!pip install -q -U google-genai
from google import genai
from getpass import getpass

api_key = getpass("Enter your Gemini API key: ")
client = genai.Client(api_key=api_key)

prompts = [
    "Write five lines about Artificial Intelligence.",
    "Explain the benefits of Python programming.",
    "Write a short paragraph about renewable energy."
]

for i, prompt in enumerate(prompts, start=1):
    print(f"\nPROMPT {i}: {prompt}")
    try:
        response = client.models.generate_content(
            model="gemini-1.5-flash",
            contents=prompt
        )
        print("GENERATED OUTPUT:")
        print(response.text)
    except Exception as e:
        print("Error:", e)

print("\nText generation experiment completed.")
```

```
Enter your Gemini API key: ••••••••••••••••••
PROMPT 1: Write five lines about Artificial Intelligence.
GENERATED OUTPUT:
1. Artificial Intelligence enables machines to perform tasks that require human intelligence.
2. It can learn from data and improve over time.
3. AI is used in healthcare, education, finance and many other fields.
4. It helps automate repetitive tasks and increases efficiency.
5. AI is transforming the way we live and work.

PROMPT 2: Explain the benefits of Python programming.
GENERATED OUTPUT:
• Easy to learn and has simple, readable syntax.
• Large ecosystem of libraries and frameworks.
• Widely used in web development, data science, automation and AI.
• Supports cross-platform development.
• Great community support and resources.

PROMPT 3: Write a short paragraph about renewable energy.
GENERATED OUTPUT:
Renewable energy comes from natural sources like sunlight, wind, water and geothermal heat. It helps reduce
pollution, lowers dependence on fossil fuels and provides a sustainable way to meet our energy needs. By using
renewable energy, we can build a cleaner and healthier environment for future generations.

Text generation experiment completed.
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6. Result

The Python program demonstrates text generation using the Google Gemini LLM API. It accepts multiple user prompts, sends them to the Gemini model, and displays the generated responses in Google Colab.